Design vs. Landscape — Gap Analysis

Deep Research Agent · our design document vs. the 2026 landscape report

2026-06-10

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Purpose

Compare our design document (`docs/superpowers/specs/2026-06-09-deep-research-agent-design.md`) against the landscape report (`docs/research/deep-research-agent-landscape-report.md`) to (a) surface ideas worth adopting and (b) record where we are genuinely differentiated. A backlog of candidate improvements for future experiments — not a commitment.

Caveat: the landscape report was compiled alongside our design (it contains a "Pattern G: Your DRA design" entry and a "Design Validation" appendix), so its praise is partly self-referential. This analysis is the *critical* read — real gaps and real divergences, not the report's self-validation.

Actionable backlog — adopt / defend / defer

#	Item	Source pattern/paper	Action	Rationale
1	**Adversarial "Skeptic"**	Pattern F (MARs, PROClaim)	**Adopt** (verifier	Moves verification from "is it

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	refutation step for `weak` / `contested` claims		slice)	supported?" to "can it be broken?"; composes with our verdict routing. We only hint at this today ("seek disconfirming evidence").
2	**Bound retrieval budget as an evidence-cited rule**	"Cited but Not Verified" (–42% acc, 2→150 calls)	**Adopt** (make explicit)	External validation of lean k=5 / max_iters≈3; promote the cap from a termination condition to a first-class design rule.
3	**Split generator role: claim-extractor vs. report-composer**	our generator-comparison finding	**Adopt** (design clarification)	Lean model wins claim extraction (faithfulness); a stronger model may still help final prose composition. The landscape blurs these.
4	**Step/process-level faithfulness signal**	VERITAS, HiPRAG, PRA	**Defer** (RL ext. + eval)	"Faithfulness ≠ correctness" — endpoint-only entailment misses unfaithful reasoning. Adds a reward/eval axis.
5	**On-demand retrieval** (skip search when a sub-q needs none)	Self-RAG `[Retrieval]` token	**Defer**	Retrieval-side cousin of "decompose only when needed"; efficiency win, not correctness.
6	**Search Intensity Scaling** (learned escalation of retrieval depth)	DeepDiver (NeurIPS)	**Defer** (RL appendix)	The learnable form of our rule-based k/maxChars escalation; design-only until training is in scope.

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7	**Multi-perspective decomposition** for breadth queries	STORM	**Defer**	Improves coverage on "map the landscape"-type prompts (our tc4); decomposer enhancement.
8	**Judge-time compute scaling**	Verdict (2502.18018)	**Defer** (eval harness)	Raises judge-panel reliability when we build the offline eval.
9	**LangGraph for the Phase-3 loop**	landscape §3.1/§7.2 (strong rec)	**Defend or adopt** (decide)	Design names it; we leaned hand-rolled. Must justify either way ("what did the framework buy / cost?").
10	**Source credibility scoring**	tarun7r/deep-research-agent	**Defer**	`reliability_note` exists in the schema but isn't actively scored.

Where we are differentiated / ahead

Per the landscape's own §6 (Gaps) and validation appendix — these are the field's *open* problems, and our design addresses them:

- **The unified loop** — verification + calibrated confidence + adaptive retrieval *together* (3/3; Self-RAG, the closest, is 2/3). "Almost no deployed system" combines all three.
- **Calibrated per-claim confidence** (ECE + temperature/isotonic) — "none deployed, only academic."
- **Verbatim evidence, never summarized** · **enforced model independence** (`assert_independence()`) · **statistical rigor** (Cohen's κ / McNemar / Holm) — each a landscape gap with ~no OSS precedent.
- **Claim ledger with epistemic status** (supported/contested/insufficient; nothing silently dropped).
- **`expected_behavior`-labeled DRA-Bench** — grades *whether the system did the right thing* (clarify / report-insufficient / flag-contested), distinguishing plausible from correct.

Empirical results this session that go beyond the report's generic advice:

- **We contradicted the landscape's generator assumption with data.** Its §3.2 recommends a *big* model as generator for "best long-form quality." Our 3-judge-panel experiment found the opposite for *grounded claim extraction*: lean **Llama 4 Maverick (17B)** beats **Mistral Large 3 (675B)** on grounding (unanimous) — the big model over-reaches from parametric memory. Insight: the "generator" splits into claim-*extractor* (lean wins) vs. report-*composer* (big may help).
- **We measured retrieval depth/budget** (cumulative coverage → k=5) and excerpt budget (→ 2k) rather than asserting them; surfaced that Exa exposes no usable relevance score (rank-only).

Divergences to decide consciously

- **LangGraph** — landscape strongly recommends; we leaned hand-rolled. Decide for Phase 3 and be ready to defend.
- **Model instances** — landscape suggests generator=Claude, verifier=GPT/Gemini; we use generator=Llama, verifier=Mistral, panel=GPT-OSS/Opus/DeepSeek. Same *principle* (family diversity), different instances — and ours are now empirically grounded, a stronger position.

Top picks if we run another round

If/when we revisit: **(1) adversarial Skeptic step** and **(3) generator-role split** are the highest-leverage, lowest-cost adoptions that strengthen the core thesis (verifiable grounding) without new infrastructure. **(9) the LangGraph decision** should be settled before Phase 3 starts.